

ROSETOWN EAST, SK, Canada

WMO: 715100

Lat: 51.570N

Lon: 107.920W

Elev: 586

StdP: 94.48

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-30.2	-37.4	0.1	-33.4	-33.8	0.2	-30.2	12.9	-3.0	11.5	-6.6	3.0	260	0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	30.9	17.6	28.7	17.0	26.8	16.6	20.3	26.4	19.1	25.2	17.9	24.2	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	14.2	23.3	16.9	13.0	21.7	15.7	11.9	20.4	61.2	26.4	56.8	25.1	53.0	24.3	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.2	9.0	DB	-37.0	34.7	3.2	2.1	-39.3	36.2	-41.2	37.4	-42.9	38.6	-45.3	40.1	(4)
(5)			WB	-37.0	23.3	3.1	1.8	-39.3	24.6	-41.1	25.7	-42.8	26.7	-45.1	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-14.4	-12.8	-5.6	3.9	10.6	15.4	18.0	17.1	12.1	4.1	-4.9	-11.8
	DBStd	13.13	8.74	7.98	8.10	5.67	4.61	3.53	3.05	3.50	4.68	5.07	6.87	7.95
	HDD10.0	3470	756	637	485	193	49	3	0	1	30	192	447	677
	HDD18.3	5784	1014	871	743	434	243	101	44	66	194	441	697	936
	CDD10.0	809	0	0	0	9	67	164	247	220	92	9	0	0
	CDD18.3	80	0	0	0	0	2	12	33	27	6	0	0	0
	CDH23.3	1477	0	0	0	8	100	214	471	497	180	6	0	0
	CDH26.7	452	0	0	0	1	21	55	141	173	60	1	0	0
Wind	WSAvg	4.2	4.1	4.0	4.4	4.8	4.9	4.5	3.6	3.6	4.1	4.3	4.1	3.9
Precipitation	PrecAvg	377	17	11	15	26	42	78	63	40	28	18	15	13
	PrecMax	756	52	23	39	131	119	149	171	99	79	53	40	25
	PrecMin	166	3	3	2	0	5	19	15	0	0	5	1	3
	PrecStd	119	11	6	9	26	30	36	40	24	22	12	11	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	4.4	14.1	24.0	29.4	31.4	33.1	33.8	31.7	23.5	14.1	6.0
		MCWB	2.5	2.3	7.6	11.4	14.6	17.5	19.2	18.1	16.4	12.7	8.1	3.0
	2%	DB	2.8	2.3	9.9	20.2	26.2	28.0	30.1	31.0	28.3	18.9	9.2	3.6
		MCWB	1.2	0.7	5.8	9.8	13.2	16.2	19.1	17.1	14.9	10.3	5.3	1.5
	5%	DB	0.7	0.6	6.5	17.1	23.5	25.8	27.9	28.4	25.0	15.5	6.2	1.6
		MCWB	-0.5	-0.4	3.7	8.4	12.3	15.4	18.6	17.1	13.7	8.7	3.2	0.0
	10%	DB	-1.9	-1.3	3.7	13.9	20.8	23.6	25.9	26.1	21.8	12.6	3.8	-1.0
		MCWB	-2.7	-2.1	2.0	6.8	11.0	14.5	18.0	16.6	12.5	7.3	1.9	-2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.4	8.5	12.3	16.0	20.1	22.7	21.1	18.0	13.8	8.6	3.3
		MCDB	4.2	4.0	13.5	21.9	24.9	26.1	28.5	27.5	28.9	20.3	13.6	5.4
	2%	WB	1.3	0.9	5.8	10.4	14.4	18.2	21.0	19.5	15.9	11.0	5.5	1.6
		MCDB	2.7	2.1	9.5	18.4	23.6	24.6	26.9	26.2	25.4	16.9	8.9	3.5
	5%	WB	-0.4	-0.4	3.8	8.9	13.0	16.8	19.8	18.3	14.4	9.3	3.6	0.0
		MCDB	0.7	0.6	6.5	15.4	21.6	22.7	25.5	25.3	22.9	14.6	5.7	1.5
	10%	WB	-2.7	-2.1	2.0	7.4	11.7	15.7	18.6	17.2	13.1	7.8	2.0	-2.0
		MCDB	-2.0	-1.3	3.6	13.1	19.6	21.4	24.3	24.0	20.8	12.3	3.7	-1.0
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.7	10.2	12.9	14.5	12.3	13.4	14.6	14.6	12.7	9.9	9.6
		MCDBR	11.2	11.7	12.8	18.8	19.3	16.6	17.1	19.4	20.9	17.8	13.3	11.6
		MCWBR	10.1	10.5	8.9	10.0	9.2	7.5	8.3	8.4	9.3	10.2	9.6	9.7
	5% WB	MCDBR	11.2	11.5	12.2	16.9	16.6	13.3	14.8	15.7	18.3	16.5	12.4	11.6
		MCWBR	10.2	10.4	8.7	9.6	8.4	6.8	8.2	8.0	8.7	9.9	9.1	10.0
Clear-Sky Solar Irradiance	taub	0.264	0.266	0.285	0.349	0.370	0.373	0.377	0.386	0.333	0.289	0.260	0.240	
	taud	2.281	2.344	2.448	2.310	2.299	2.346	2.356	2.297	2.425	2.529	2.474	2.380	
	Ebn at Noon	755	874	924	888	883	881	870	840	855	842	778	745	
	Edh at Noon	64	80	89	115	123	118	115	116	91	67	53	49	
All-Sky Solar Radiation	RadAvg	1.14	2.17	3.55	4.79	5.88	6.01	6.46	5.21	3.79	2.27	1.15	0.82	
	RadStd	0.12	0.17	0.28	0.43	0.43	0.46	0.33	0.39	0.35	0.31	0.13	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page